

LTA analysis - Descriptive Stats

Annie Green Howard

Reading in dataset

Reading in the combined years datasets

```
combine_yrs <- read.table("Original Data/lca_combined_years.dat",
                          header = TRUE, sep = ",", skip = 2)

#Changing all values of -9999 to missing
combine_yrs[combine_yrs==-9999] <- NA
```

Adding in variable names to file

There appear to be a total of 16,947 observations with 78 variables.

It's not completely clear in the published paper there were 8,702 individuals in 2009 and 7,929 in 2019 - for a total sample size of 16,631 - but this doesn't seem to allow me to link the same individuals over time.

Recreating descriptive tables

We could potentially assume group is year given group = 1 for 7,929 individuals, which matches the sample size in published paper for 2019. However, there are a total of 9,018 with a value of group = 0 - which is larger than the 8,702 individuals in the descriptive table for 2009.

```
table(combine_yrs$group)
```

0	1
9018	7929

Looking at the 2009 dataset and getting it to be similar to 2019 dataset

The 2009 dataset does include closer to the true number, including 8,699 individuals. Same sample size for both 1.24.23 version and the v2 version. These datasets both include indexno variable. We cannot seem to get to exactly 8,702 individuals, in Table 1 in published paper, was there exclusion criteria we care about there where we might also want to include them in the descriptive table for LTA work?

```
#Reading in the version 2 datasets for 2009
##but this seems to be an older version, dated 11/22/2023
lca_pcp_2009v2      <- read.table("Original Data/lca_pcp_2009v2.dat",
                                header = TRUE, sep = ",", skip = 2)
#Changing all values of -9999 to missing
lca_pcp_2009v2[lca_pcp_2009v2== -9999] <- NA

#Reading in the datasets for 2009 and 2019 as of 1.24.23
lca_pcp_2009_12423 <- read.table("Original Data/lca_pcp_2009_1.24.23.dat",
                                header = TRUE, sep = ",", skip = 2)
#Changing all values of -9999 to missing
lca_pcp_2009_12423[lca_pcp_2009_12423== -9999] <- NA
```

Now we need to format the variables to be categorical variables

```
lca_pcp_2009_12423 <- lca_pcp_2009_12423 %>%
  mutate(specialty = case_when(
    group_fam    == 1 ~ "Family Medicine",
    group_int    == 1 ~ "General Internal Medicine",
    group_obgyn  == 1 ~ "General Obstetrics and Gynecology",
    group_ped    == 1 ~ "General Pediatrics",
    group_ger    == 1 ~ "Geriatrics",
    TRUE         ~ "Missing"
  ),
  employment_setting = case_when(
    setting_amb  == 1 ~ "Ambulatory Care",
    setting_hosp == 1 ~ "Hospital",
    setting_medsch == 1 ~ "Medical School/Parent University",
    setting_nurs == 1 ~ "Nursing Home/Extended Care",
    setting_other == 1 ~ "Other",
    setting_solo == 1 ~ "Solo practioner",
    setting_missing == 1 ~ "Missing",
    TRUE         ~ "Missing Value"
  ),
```

```

urban_rural = case_when(
  ruca_urban == 1 ~ "Urban",
  ruca_lrural == 1 ~ "Large Rural",
  ruca_srural == 1 ~ "Small Rural",
  TRUE ~ "Missing"
),
race_ethnicity = case_when(
  white == 1 ~ "Non-Hispanic White",
  black == 1 ~ "Non-Hispanic Black",
  american_native == 1 ~ "American Indian/Alaskan Native",
  asian_or_pi == 1 ~ "Asian/Pacific Islander",
  hispanicrace == 1 ~ "Hispanic",
  other_race == 1 ~ "Other",
  race_unknown == 1 ~ "Unknown",
  TRUE ~ "Missing"
),
generational = case_when(
  gen_x == 1 ~ "Gen X (1965-1980)",
  gen_y == 1 ~ "Gen Y (1981-2000)",
  gen_boom_lead == 1 ~ "Baby Boomer, Leading Edge (1946-1954)",
  gen_boom_trail == 1 ~ "Baby Boomer, Trailing Edge (1955-1964)",
  gen_ww2 == 1 ~ "WWII (1928-1945)",
  TRUE ~ "Missing"
))

```

Specialty

We do note that 3 individuals, out of the 8,699 are missing specialty. Although none are mentioned in Table 1 as missing specialty.

```
table(lca_pcp_2009_12423$specialty)
```

Family Medicine	General Internal Medicine
2826	3071
General Obstetrics and Gynecology	General Pediatrics
1127	1630
Geriatrics	Missing
42	3

Employment Setting

We do have 550 individuals missing employment setting which aligns with Table 1, where we see some degree of missingness for employment setting. Do we drop these individuals from the analysis? We could leave them in but there are two different approaches - it's not super clear what was used before but we could essentially use maximum likelihood methods - leave them as missing - or create missingness categories. I think it depends on why think people are missing.

```
table(lca_pcp_2009_12423$employment_setting)
```

Ambulatory Care	Hospital
4604	1750
Medical School/Parent University	Missing
648	550
Nursing Home/Extended Care	Other
71	113
Solo practioner	
963	

Practice Characteristics

While other variables are mutually exclusive providing routine obstetric care and providing routine prenatal care appear to be not mutually exclusive.

```
table(lca_pcp_2009_12423$pregnancy, lca_pcp_2009_12423$obstetrics)
```

	0	1
0	7060	20
1	262	848

Average patient-care hours per week

Check on number of missing values for average patient-care hours per week - we have a total of 626 individuals missing.

```
sum(is.na(lca_pcp_2009_12423$patient_care_hours))
```

```
[1] 626
```

Looking at descriptive statistics by patient care

```
#Mean for patient care hours
mean(lca_pcp_2009_12423$patient_care_hours, na.rm=TRUE)
```

```
[1] 40.71436
```

```
#Mean for Standardized values of Y2009_patient_care_hours
mean(lca_pcp_2009_12423$zhours, na.rm=TRUE)
```

```
[1] 0.0004237322
```

Urban/rural status

```
table(lca_pcp_2009_12423$urban_rural)
```

Large Rural	Small Rural	Urban
1202	497	7000

Female

```
table(lca_pcp_2009_12423$female)
```

0	1
5266	3433

Racial/ethnic identity

Race/ethnicity does appear to be already classified as mutually exclusive

```
table(lca_pcp_2009_12423$white,lca_pcp_2009_12423$hispanicrace)
```

	0	1
0	2139	304
1	6256	0

Checking with other race/ethnicity groups.

```
table(lca_pcp_2009_12423$white,lca_pcp_2009_12423$black)
```

```

      0      1
0 1518   925
1 6256     0

```

```
table(lca_pcp_2009_12423$race_ethnicity)
```

```

American Indian/Alaskan Native      Asian/Pacific Islander
                        60                        859
                Hispanic      Non-Hispanic Black
                304                        925
        Non-Hispanic White                        Other
                6256                        257
                Unknown
                38

```

Generational Cohort

```
table(lca_pcp_2009_12423$generational)
```

```

Baby Boomer, Leading Edge (1946-1954)  Baby Boomer, Trailing Edge (1955-1964)
                        1700                        2461
        Gen X (1965-1980)                        Gen Y (1981-2000)
                        3867                        9
        WWII (1928-1945)
                        662

```

Average years of practice

There does appear to be 18 individuals missing practice years

```
sum(is.na(lca_pcp_2009_12423$practice_yrs))
```

```
[1] 18
```

Now looking at average years of practice among those with non-missing values

```
mean(lca_pcp_2009_12423$practice_yrs, na.rm = TRUE)
```

```
[1] 19.95358
```

Average age

No individuals missing values for age

```
sum(is.na(lca_pcp_2009_12423$age))
```

```
[1] 0
```

Now looking at average years of practice among those with non-missing values

```
mean(lca_pcp_2009_12423$age, na.rm = TRUE)
```

```
[1] 46.50213
```

Medical school

Seem we consider IMG medical school vs. other

```
table(lca_pcp_2009_12423$school_img)
```

```
  0    1  
7196 1455
```

Looking at the 2019 dataset

Note that the variables in the 2009 dataset are the exact same variables in the 2019 dataset.

There are a total of 7,926 individuals in the 2019 dataset.

```
lca_pcp2019_12423 <- read.table("Original Data/lca_pcp2019_1.24.23.dat",  
                               header = TRUE, sep = ",", skip = 2)  
#Changing all values of -9999 to missing  
lca_pcp2019_12423[lca_pcp2019_12423==-9999] <- NA  
names(lca_pcp2019_12423) <- vars_2009
```

Now we need to format the variables to be categorical variables

```
lca_pcp2019_12423 <- lca_pcp2019_12423 %>%
  mutate(specialty = case_when(
    group_fam == 1 ~ "Family Medicine",
    group_int == 1 ~ "General Internal Medicine",
    group_obgyn == 1 ~ "General Obstetrics and Gynecology",
    group_ped == 1 ~ "General Pediatrics",
    group_ger == 1 ~ "Geriatrics",
    TRUE ~ "Missing"
  ),
  employment_setting = case_when(
    setting_amb == 1 ~ "Ambulatory Care",
    setting_hosp == 1 ~ "Hospital",
    setting_medsch == 1 ~ "Medical School/Parent University",
    setting_nurs == 1 ~ "Nursing Home/Extended Care",
    setting_other == 1 ~ "Other",
    setting_solo == 1 ~ "Solo practioner",
    setting_missing == 1 ~ "Missing",
    TRUE ~ "Missing Value"
  ),
  urban_rural = case_when(
    ruca_urban == 1 ~ "Urban",
    ruca_lrural == 1 ~ "Large Rural",
    ruca_srural == 1 ~ "Small Rural",
    TRUE ~ "Missing"
  ),
  race_ethnicity = case_when(
    white == 1 ~ "Non-Hispanic White",
    black == 1 ~ "Non-Hispanic Black",
    american_native == 1 ~ "American Indian/Alaskan Native",
    asian_or_pi == 1 ~ "Asian/Pacific Islander",
    hispanicrace == 1 ~ "Hispanic",
    other_race == 1 ~ "Other",
    race_unknown == 1 ~ "Unknown",
    TRUE ~ "Missing"
  ),
  generational = case_when(
    gen_x == 1 ~ "Gen X (1965-1980)",
    gen_y == 1 ~ "Gen Y (1981-2000)",
    gen_boom_lead == 1 ~ "Baby Boomer, Leading Edge (1946-1954)",
    gen_boom_trail == 1 ~ "Baby Boomer, Trailing Edge (1955-1964)",
    gen_ww2 == 1 ~ "WWII (1928-1945)",
```

```
TRUE ~ "Missing"
))
```

Specialty

We do have no individuals missing specialty in 2019, which aligns with descriptive table.

```
table(lca_pcp2019_12423$specialty)
```

Family Medicine	General Internal Medicine
2825	2062
General Obstetrics and Gynecology	General Pediatrics
1160	1629
Geriatrics	
250	

Employment Setting

We do have 572 individuals missing employment setting which aligns with Table 1, where we see some degree of missingness for employment setting

```
table(lca_pcp2019_12423$employment_setting)
```

Ambulatory Care	Hospital
4305	1443
Medical School/Parent University	Missing
240	572
Nursing Home/Extended Care	Other
123	678
Solo practioner	
565	

How would we want to handle missing categories? Drop them from the analysis?

Practice Characteristics

While other variables are mutually exclusive providing routine obstetric care and providing routine prenatal care appear to be not mutually exclusive.

```
table(lca_pcp2019_12423$pregnancy, lca_pcp2019_12423$obstetrics)
```

```

      0      1
0 6430    35
1  251   916

```

Average patient-care hours per week

Check on number of missing values for average patient-care hours per week - we have a total of 698 individuals missing.

```
sum(is.na(lca_pcp2019_12423$patient_care_hours))
```

```
[1] 698
```

Looking at descriptive statistics by patient care

```

#Mean for patient care hours
mean(lca_pcp2019_12423$patient_care_hours, na.rm=TRUE)

```

```
[1] 38.38046
```

```

#Mean for Standardized values of Y2009_patient_care_hours
mean(lca_pcp2019_12423$zhours, na.rm=TRUE)

```

```
[1] -0.0002234069
```

Urban/rural status

We do have 11 individuals missing Urban/Rural in 2019

```
table(lca_pcp2019_12423$urban_rural)
```

```

Large Rural      Missing Small Rural      Urban
      944           11          367      6604

```

Female

```
table(lca_pcp2019_12423$female)
```

```

0      1
3957 3969

```

Racial/ethnic identity

Can't seem to identify the 3 individuals missing - unless they are captured within the 267 unknown?

```
table(lca_pcp2019_12423$race_ethnicity)
```

American Indian/Alaskan Native	Asian/Pacific Islander
48	835
Hispanic	Non-Hispanic Black
321	899
Non-Hispanic White	Other
5349	207
Unknown	
267	

Generational Cohort

```
table(lca_pcp2019_12423$generational)
```

Baby Boomer, Leading Edge (1946-1954)	Baby Boomer, Trailing Edge (1955-1964)
911	1790
Gen X (1965-1980)	Gen Y (1981-2000)
3340	1686
WWII (1928-1945)	
199	

Average years of practice

There does appear to be 4 individuals missing practice years

```
sum(is.na(lca_pcp2019_12423$practice_yrs))
```

```
[1] 4
```

Now looking at average years of practice among those with non-missing values

```
mean(lca_pcp2019_12423$practice_yrs, na.rm = TRUE)
```

```
[1] 22.46541
```

Average age

No individuals missing values for age

```
sum(is.na(lca_pcp2019_12423$age))
```

```
[1] 0
```

Now looking at average years of practice among those with non-missing values

```
mean(lca_pcp2019_12423$age, na.rm = TRUE)
```

```
[1] 49.22117
```

Medical school

Seem we consider IMG medical school vs. other

```
table(lca_pcp2019_12423$school_img)
```

```
  0    1  
6709 1214
```

Creating combined datasets

Changing variable names to __2009 or __2019 depending on what dataset they are from

Now merging the dataset keeping only individuals in both datasets

```
merged_lta <- merge(lca_pcp_2009_12423, lca_pcp2019_12423, by = "indexno")
```

This results in a total of 4,614 individuals. Could we look at descriptive statistics for combinations of variables across time

In particular we are not sure about specialty, given questions have changed over time so this shifted broad patterns of what specialties show up. Not necessarily a change of practice, but a change in how they reported their practice.

Descriptive stats for people in 2009 dataset only

Descriptive stats for people in 2019 dataset only

Getting from the 2009

Now looking at descriptive stats for speciality across years

```
table(merged_lta$specialty_2009,merged_lta$specialty_2019)
```

	Family Medicine	General Internal Medicine
Family Medicine	1619	3
General Internal Medicine	22	1156
General Obstetrics and Gynecology	8	1
General Pediatrics	6	6
Geriatrics	1	6
Missing	0	1

	General Obstetrics and Gynecology
Family Medicine	4
General Internal Medicine	0
General Obstetrics and Gynecology	660
General Pediatrics	0
Geriatrics	0
Missing	0

	General Pediatrics	Geriatrics
Family Medicine	9	77
General Internal Medicine	40	50

General Obstetrics and Gynecology	0	0
General Pediatrics	919	1
Geriatrics	0	23
Missing	1	1

```
table(merged_lta$employment_setting_2009,merged_lta$employment_setting_2019)
```

	Ambulatory Care	Hospital
Ambulatory Care	2320	208
Hospital	135	204
Medical School/Parent University	26	31
Missing	132	30
Nursing Home/Extended Care	2	0
Other	14	5
Solo practioner	214	24

	Medical School/Parent University	Missing
Ambulatory Care	15	124
Hospital	10	25
Medical School/Parent University	102	3
Missing	9	15
Nursing Home/Extended Care	0	1
Other	0	2
Solo practioner	2	19

	Nursing Home/Extended Care	Other
Ambulatory Care	29	207
Hospital	8	86
Medical School/Parent University	1	10
Missing	3	25
Nursing Home/Extended Care	32	7
Other	0	10
Solo practioner	4	37

	Solo practioner
Ambulatory Care	133
Hospital	21
Medical School/Parent University	1
Missing	23
Nursing Home/Extended Care	0
Other	2

```
table(merged_lta$urban_rural_2009,merged_lta$urban_rural_2019)
```

	Large Rural	Missing	Small Rural	Urban
Large Rural	512	0	21	108
Small Rural	22	0	185	46
Urban	62	1	31	3626